6 **(a)** By reference to the photoelectric effect, explain

(i) why the existence of a very short emission time provide evidence for the particulate nature of electromagnetic radiation, as opposed to wave theory,

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.... [2]

(ii) what is meant by the *work function* of a surface,

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.... [1]

(iii) why, even when the incident light is monochromatic, the emitted electrons have a range of kinetic energy up to a maximum value.

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.... [2]

(b) Two beams of monochromatic light have similar intensities. The light in one beam has wavelength 350 nm and the light in the other beam has wavelength 700 nm.

The two beams are incident separately on three different metal surfaces. The work function of each of these surfaces is shown in Fig. 6.1.

metal	work function/ eV
tungsten	4.5
magnesium	3.8
potassium	2.3

Fig. 6.1

State which combination, if any, of monochromatic light and metal surface could give rise to photoelectric emission. Give a quantitative explanation of your answer.

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7 Read the passage below and answer the questions that follow.